

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 02

Create and Secure an AWS IAM Administrator Account

AWS IAM | Group-Based Permissions | Privileged Access

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 8, 2026
STATUS	Complete

Executive Summary

Created a dedicated AWS IAM administrator account to eliminate routine use of the AWS root identity. Administrative privileges were assigned through an IAM group, the user was protected with a strong password and MFA, and the final configuration was validated with redacted screenshots.

Business Problem

Using the root identity for day-to-day administration concentrates unnecessary risk in the most powerful account credential. The environment needed a separate privileged identity that could perform administrative work while keeping root access reserved for exceptional tasks.

Objectives

- Reduce routine dependence on the AWS root identity.
- Create a dedicated administrative IAM user.
- Manage administrator permissions through an IAM group.
- Protect the privileged user with MFA.
- Verify that the user has no unnecessary long-term access keys.

Environment

Cloud Provider	Amazon Web Services (AWS)
Service	AWS Identity and Access Management (IAM)
User Type	Dedicated IAM administrator
Permission Source	Administrators group
Policy	AWS managed AdministratorAccess
Authentication	Strong password + virtual MFA
Access Keys	None

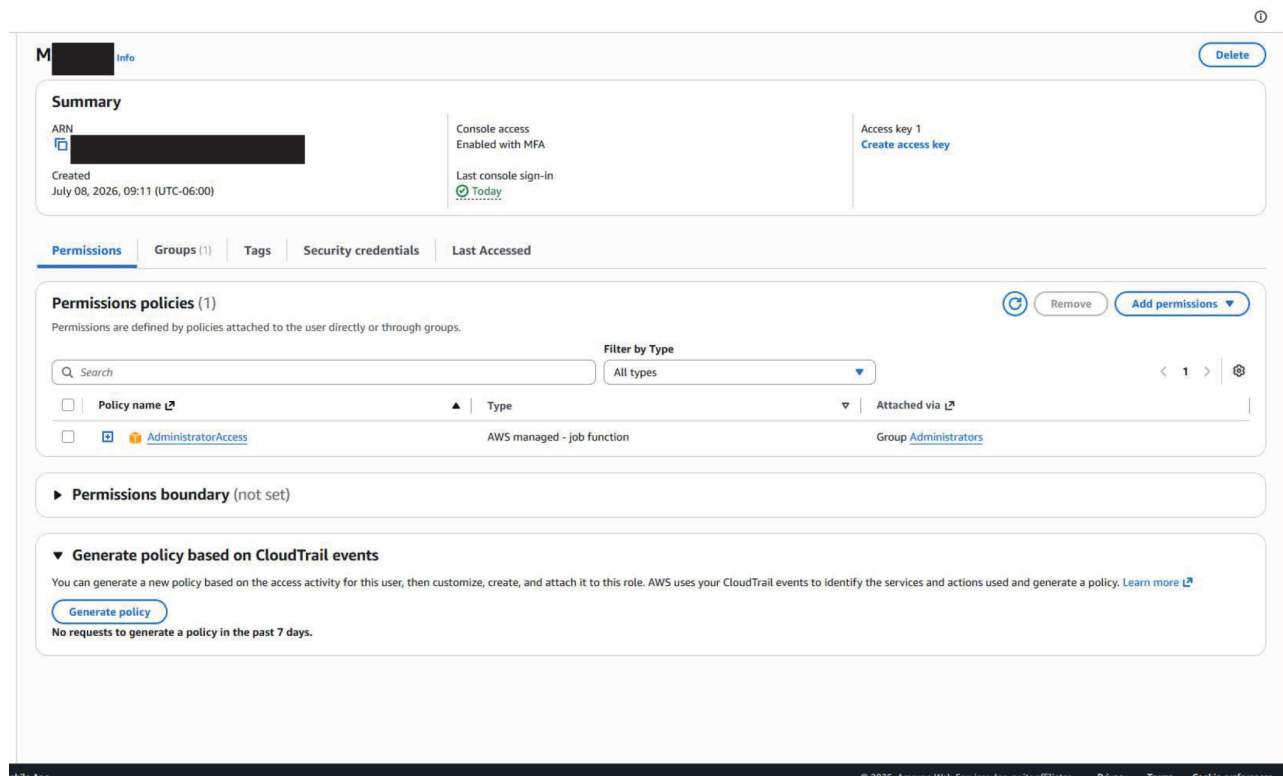
Implementation Summary

1. Signed in with the secured root identity for initial account configuration.
2. Created an IAM group named Administrators.
3. Attached the AWS-managed AdministratorAccess policy to the group.

4. Created a dedicated IAM administrator with console access and a strong password.
5. Assigned the user to the Administrators group so permissions were inherited through group membership.
6. Signed in as the new IAM administrator and enabled MFA.
7. Validated the final permission path, MFA status, and absence of access keys.

Evidence

Figure 1 - Administrator Permission Inheritance



The IAM user permissions page shows the AdministratorAccess managed policy attached through the Administrators group rather than directly to the user. This demonstrates centralized permission management.

Figure 2 - Administrator Security Credentials and MFA

The screenshot shows the 'Security credentials' tab for an AWS IAM user. The 'Summary' section indicates console access is enabled with MFA, the last console sign-in was today, and there is one access key. The 'Console sign-in' section shows the console sign-in link is redacted and the console password was updated 3 hours ago. The 'Multi-factor authentication (MFA)' section shows one virtual MFA device assigned, with its identifier redacted. The 'Access keys' section shows zero access keys. The 'API keys' section shows zero API keys.

The security credentials page confirms console access is enabled with MFA and shows zero access keys. Sensitive identifiers and sign-in details were redacted before publication.

Security Analysis

Moving routine administration away from the root identity reduces dependence on the most powerful account credential and improves separation between account ownership and daily operations.

Permissions were inherited through the Administrators group instead of being attached directly to the user. Group-based authorization is easier to review, scale, and maintain than per-user policy assignment.

MFA protects the privileged console account from password-only compromise, while the absence of access keys avoids unnecessary long-term programmatic credentials.

Lessons Learned

Identity in AWS is both an authentication and authorization problem. A secure administrative identity requires strong sign-in controls and a clear permission model.

Using groups makes the source of permissions easier to identify. The screenshot shows AdministratorAccess attached through the Administrators group, which improves traceability.

Production Improvements

- Replace routine IAM-user access with federation or IAM Identity Center for workforce access.

- Retain a carefully controlled break-glass access path and test it periodically.
- Require phishing-resistant MFA for highly privileged identities.
- Monitor privileged sign-ins and permission changes with CloudTrail and alerting.

Skills Demonstrated

IAM user creation	MFA	Group-based access
IAM group management	Privileged access management	Policy inheritance
AWS managed policies	Credential hygiene	Access review

Resume Bullet

RESUME	Implemented secure AWS administrative access by creating a dedicated IAM administrator, assigning AdministratorAccess through an IAM group, enforcing MFA, and avoiding long-term access keys.
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STAR Interview Story

Situation: The new AWS environment still depended on the root identity for administration.

Task: Create a safer day-to-day administrative access model.

Action: I created an Administrators group, attached the AdministratorAccess policy, created an IAM administrator, assigned the user to the group, configured a strong password, enabled MFA, and verified that no access keys were created.

Result: Routine administration could move away from the root identity while permissions remained centrally managed and protected by multi-factor authentication.

Version History

v1.0.0	July 8, 2026	Initial release. Added the Administrators group with AdministratorAccess, a dedicated IAM administrator, MFA, and validation that no access keys were present.

References

Cloud Security Lab a Week (S.L.A.W.) - Create and Secure Your First Admin User.; User-provided, redacted IAM permissions screenshot.; User-provided, redacted IAM security credentials screenshot.